Correlation Exercises

NEB Class 12 Management Mathematics

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Section A: Multiple Choice Questions (MCQs)

1. If the value of Karl Pearson's correlation coefficient is -0.85, the relationship between X and Y is:
   (a) Weak negative
   (b) Strong negative
   (c) Strong positive
   (d) No correlation

2. If $r = 0$, it means:
   (a) Perfect positive correlation
   (b) Perfect negative correlation
   (c) No linear correlation
   (d) Non-linear correlation exists

3. The range of Pearson's correlation coefficient is:
   (a) 0 to $+1$
   (b) -1 to $+1$
   (c) $-\infty$ to $+\infty$
   (d) 0 to ∞

4. Spearman's rank correlation is more suitable when:
   (a) Data are continuous
   (b) Data are nominal
   (c) Data are ordinal / ranks are given
   (d) None of the above

5. If $r = 1$, it implies:
   (a) Perfect positive correlation
   (b) Perfect negative correlation
   (c) No correlation
   (d) Weak correlation

Section B: Numerical Questions

1. Calculate Karl Pearson's coefficient of correlation:

X	1	2	3	4	5
Y	2	4	5	4	5

2. Find the correlation coefficient between X and Y:

X	10	20	30	40	50
Y	8	12	15	20	24

3. Calculate r for the following data:

X	2	4	6	8	10
Y	5	7	9	8	11

4. Calculate correlation coefficient for the following marks:

Math (X)	35	40	50	60	65	70	75	80	85	90
Stat (Y)	30	35	48	55	63	65	70	75	80	88

5. Compute Karl Pearson's correlation coefficient:

Age (X)	45	50	55	60	65	70
BP (Y)	128	130	135	138	142	150

6. Compute Spearman's rank correlation coefficient:

Economics (X)	25	28	35	30	32	36
Accounts (Y)	40	38	45	35	30	50

7. Calculate Spearman's rank correlation coefficient:

X	68	64	75	50	64	80
Y	62	58	68	45	81	60

8. Calculate Spearman's rank correlation coefficient:

Statistics	10	20	30	40	50	60	70
Math	12	18	28	35	45	55	65

9. From the following data calculate Karl Pearson's correlation coefficient:

X	1	2	3	4	5	6
Y	9	8	10	12	11	13

10. Calculate Karl Pearson's correlation coefficient:

X	10	20	30	40	50	60	70	80
Y	15	25	35	45	55	65	75	85

11. Find Spearman's rank correlation coefficient (ranks already given):

Student	**Rank in Subject A**	**Rank in Subject B**
1	1	2
2	2	1
3	3	4
4	4	3
5	5	5